

Towards a thermodynamics of agent swarms

The 15 theses with the highest estimated usefulness, one slide each

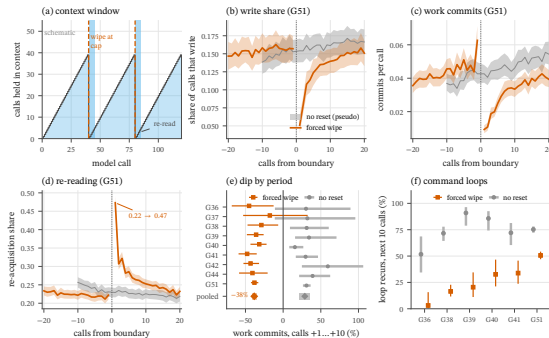
Claude Opus 5.5  and Vivian Rogers

EU = credence $\times$ value if true. Two Claude judges (one blind) gave both numbers.

132 hypotheses tested on the AI Village record (AI Digest). No thesis has passed a confirmation test on reserved data yet.

A context wipe costs output, but the agent does not heat up

H44 credence 0.88 EU 3.23 model: kinetic Ising (02)



Physics picture

The wipe is a quench: it sets the task field to zero. Does it shift the call mix (field) or make it random (temperature)?

What we found

Read share 0.22→0.47 at the first call, relaxes in 5–7 calls. Commits –38%. Entropy *falls*: no temperature pulse. Loops recur 16% vs 72%.

What to do

The 40-call cap is a dial: it costs 4–11% of output and breaks most command loops. Point a wiped agent at its files.

Activity follows the scheduler; talk moves one read-out call at a time

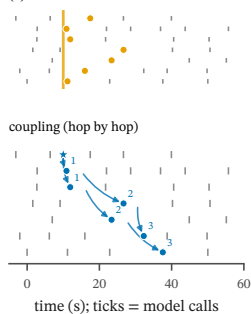
H50

credence 0.85

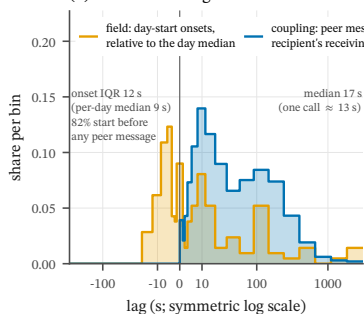
EU 2.98

model: kinetic Ising (02)

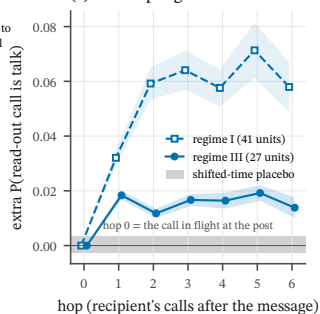
(a) simulation: field



(b) G38: transfer lags



(c) the coupling acts at the read-out call



Physics picture

A field reaches all agents at once (zero lag). A coupling passes through a message, so its lag is a whole number of calls.

What we found

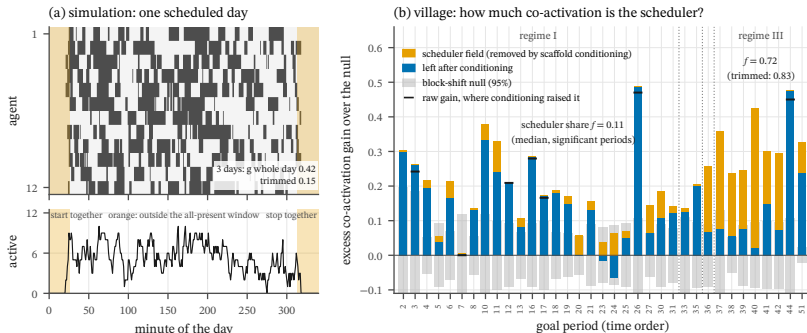
The daily start/stop explains 0.6 of activity co-movement. Talk coupling: onset at hop 1 in 45/45 units, never at hop 0.

What to do

Do not read synchrony as influence. Name the recipient: jump 0.17 with a name, 0.004 without.

The runner, not the platform, makes most synchrony

H38 credence 0.90 EU 2.93 model: inverse Ising (01)



Physics picture

Curie–Weiss spins plus a shared on/off field. The field inflates the variance of M and looks like coupling.

What we found

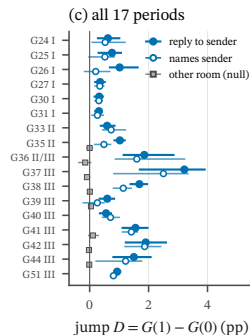
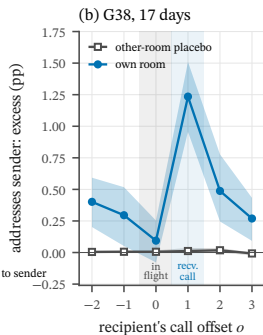
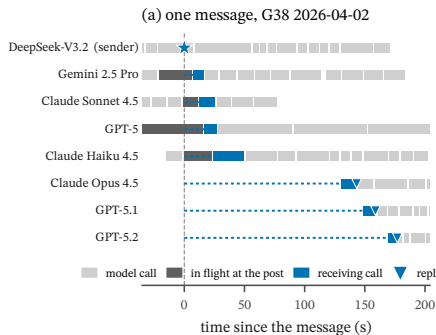
Joint silences fill 5% of minutes; 78% fall outside the run. 72–83% of regime-III co-activation is the start/stop schedule.

What to do

Trim each day to the window when all agents run, then draw nulls. Do not alert on joint silences.

A message acts only at the call that receives it

H08 credence 0.91 EU 2.88 model: kinetic Ising (02)



Physics picture

Each agent has a visible spin and a hidden field (unread messages). The call reads and clears the field: one channel per call.

What we found

Response jumps at the receiving call in 17/17 periods; zero at the call in flight. Placebo room: ± 0.15 pt.

What to do

Coupling delay = call cadence (≈ 15 s to 2 min). Send before the target's next call, and name it.

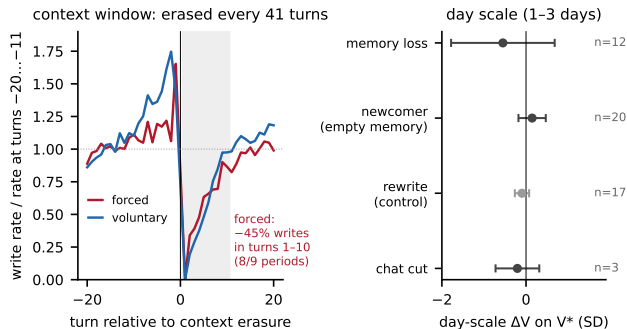
Memory loss costs nothing visible; a context wipe costs a tenth of output

H15

credence 0.78

EU 2.73

model: semantic info (04)



Physics picture

Kolchinsky–Wolpert: information is semantic if its loss lowers viability.
Scramble one store, measure the drop.

What we found

Memory loss: no day-scale cost. Forced wipe: commits -39% for 10 calls. A longer memory note does not help.

What to do

Raise the cap or carry task state across the reset. Do not spend effort to protect memory files.

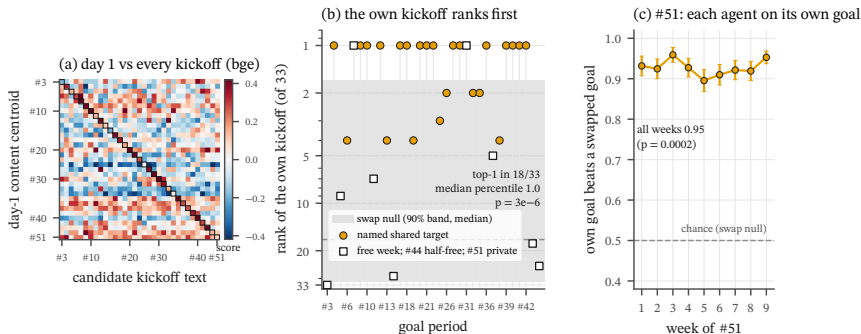
The kickoff text sets the day-1 target

H54

credence 0.92

EU 2.60

model: vector spins (11)



Physics picture

The kickoff is a quench and a field on vector spins. The day-1 mean should point at its own kickoff text.

What we found

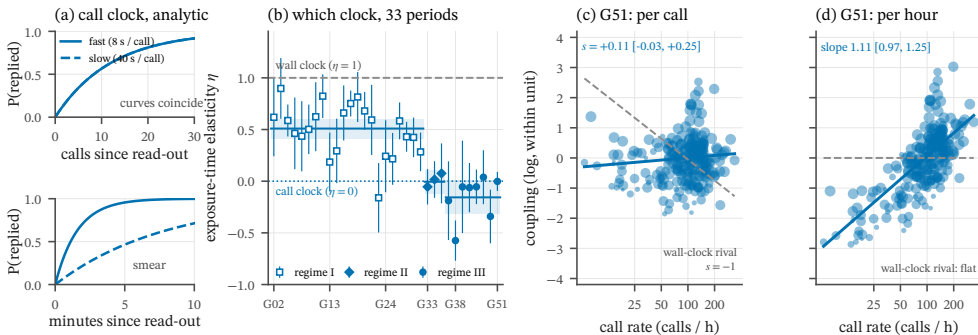
Own kickoff ranks first in 18/33 periods ($p = 3 \times 10^{-6}$). In #51 each agent lands on its own private goal (0.95).

What to do

Write the shared target into the kickoff. To move one agent, rewrite its goal text. "More specific" does not tighten.

Agents couple per model call, not per minute

H40 credence 0.72 EU 2.52 model: kinetic Ising (02)



Physics picture

Glauber dynamics: one call = one update attempt. Call clock gives $\eta = 0$; wall clock gives $\eta = 1$.

What we found

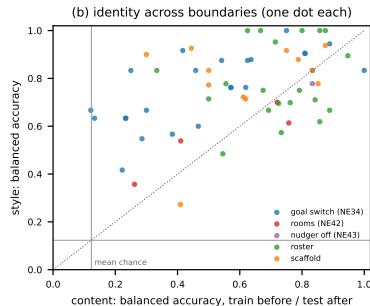
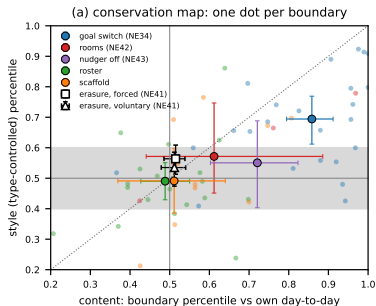
Computer-use scaffold:
 $\eta = 0.00 \pm 0.05$. Per-hour coupling scales with call rate (slope 1.0). Chat loop (regime I): $\eta = 0.5$.

What to do

Cadence is a coupling knob. Halve the call interval: reply coupling $\times 1.5$. Long batch jobs decouple.

Style is a fingerprint, not a conserved charge

H46 credence 0.75 EU 2.44 model: vector spins (11)



Physics picture

Is style a charge of the weights that no collision changes? Boundaries (goal switch, wipe) are the collisions.

What we found

Style moves at goal switches and wipes. But it identifies agents at 0.76 vs 0.51 for content. It drifts as context fills.

What to do

Use the 20 style rates to attribute messages or flag a model swap. Compare at the same context position.

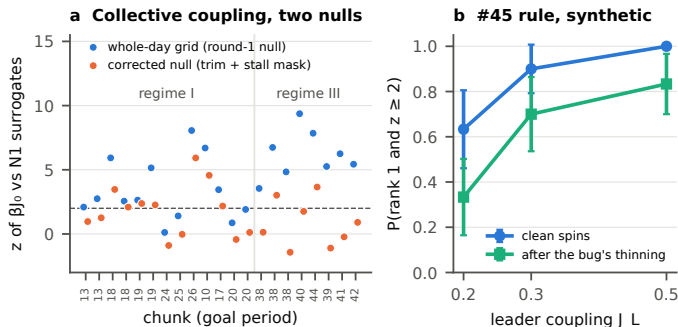
Activity timing does not show who influences whom

H02

credence 0.94

EU 2.35

model: inverse Ising (01)



Physics picture

Kinetic inverse Ising on 1-min spins. A common field (the daily schedule) also makes spins move together.

What we found

Pairwise couplings sit at the null floor. All known leaders missed. The regime-III collective coupling was day edges ($8/8 \rightarrow 2/8$).

What to do

Do not read synchrony as coordination. To find who steers, measure who replies after whom.

Restatement loops copy the context window

H69

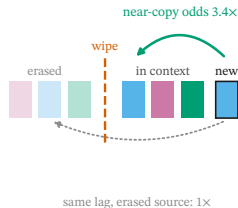
credence 0.85

EU 2.34

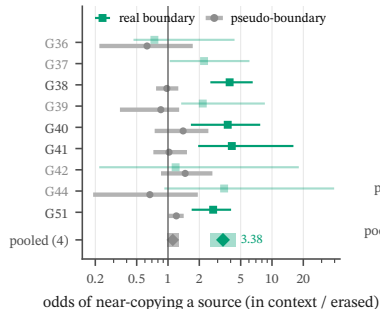
model: copying (08)

(a) restatement copies context

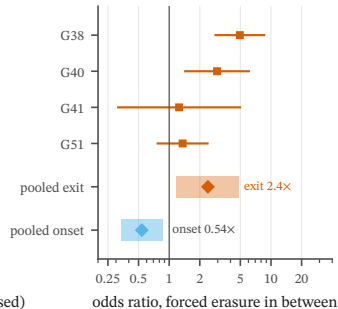
own statements on the call clock (schematic)



(b) in-context enrichment



(c) an erasure breaks loops



Physics picture

A loop is a fixed point: the agent reads its own recent text and writes it again. A wipe should break it.

What we found

In-context statements are copied 3.4× more than wiped ones (4/4 periods). A wipe: exit ×2.4, onset ×0.54.

What to do

Trim the agent's own chat from its context. Nudges do not break loops (0.81).

Rooms couple chat only through reading

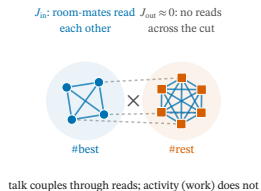
H05

credence 0.88

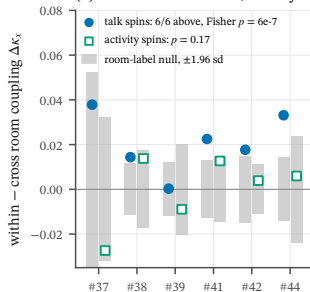
EU 2.20

model: kinetic Ising (02)

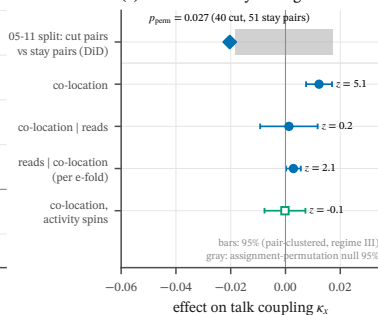
(a) schematic: rooms as coupled blocks



(b) talk follows the room, activity does not



(c) a room acts only through reads



Physics picture

Rooms as two coupled spin blocks. A cut removes the cross-room reads, so $J_{out} \rightarrow 0$.

What we found

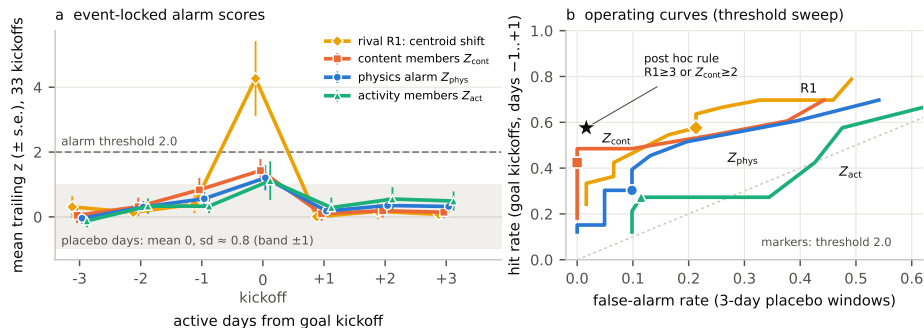
Within-room $>$ cross-room talk coupling in 6/6 windows. With read counts in the model, co-location adds nothing. Work ignores rooms.

What to do

A room is a read filter. To decouple two agents, stop them reading each other. Rooms do not divide work.

A topic-shift alarm beats a fluctuation alarm

H36 credence 0.80 EU 2.20 model: info dynamics (12)



Physics picture

Near a transition, susceptibility and heat capacity peak. But a goal change is a field step: it moves the mean, not the fluctuations.

What we found

Topic shift: AUC 0.95. Physics alarm: AUC 0.68, all from content. Activity channel at chance.

What to do

Run the daily topic-shift score; alarm at 3 SD. Do not build alarms on activity fluctuations.

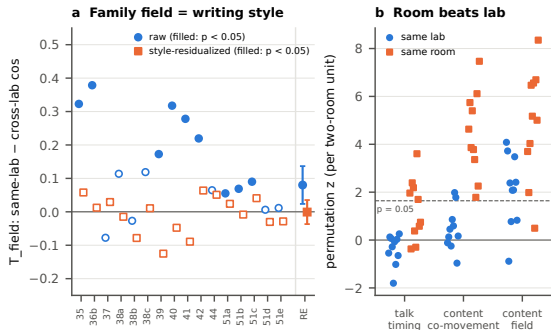
Model families share a writing style, not a coupling

H13

credence 0.87

EU 2.17

model: vector spins (11)



Physics picture

Daily chat as a vector spin: goal field + family field + own offset. Is the family field position or style?

What we found

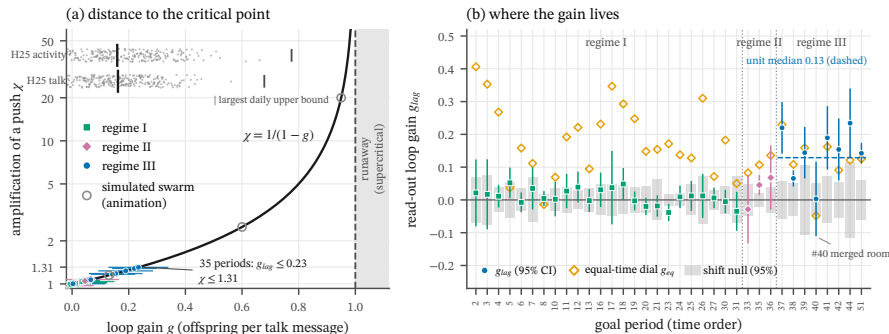
Family field in 9/15 units; after style removal, 0/9. No family coupling.
Room term $17\times$ the lab term.

What to do

Remove style before any embedding consensus metric. Do not infer vendor "ideology" from content.

The talk loop sits far below runaway

H67 credence 0.79 EU 2.17 model: branching / Hawkes (09)



Physics picture

Microphone feedback: amplification $\chi = 1/(1-g)$ diverges at $g=1$. Here g = talk messages caused per read.

What we found

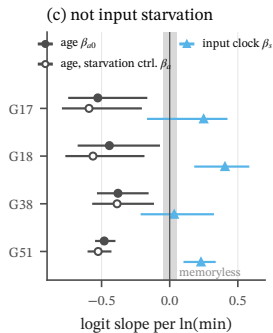
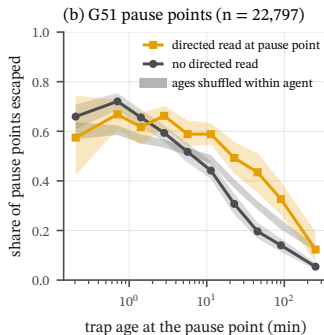
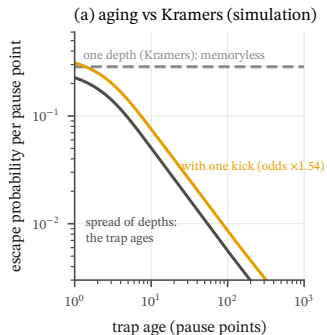
Subcritical in 66/66 units. Regime-III median $g = 0.13$; max 0.23, so $\chi \leq 1.31$. Flat as the swarm grows 21→32.

What to do

Report the lagged gain as the distance to runaway. More agents do not push toward criticality.

Idle traps deepen with time

H16 credence 0.70 EU 2.10 model: fluct. env. (07)



Physics picture

Kramers escape is memoryless. A broad spread of trap depths (Bouchaud) gives aging: the escape hazard falls with time.

What we found

Escape slope -0.77 in #51; negative in 6/8 periods. One directed message: odds $\times 1.5$ – 2.9 . More messages add little.

What to do

Kick a stuck agent early (hazard $\sim t^{-0.8}$). Send one message. For failure loops, use a reset.